Supply Chain Analysis and Optimisation

Team Lunatic Byte

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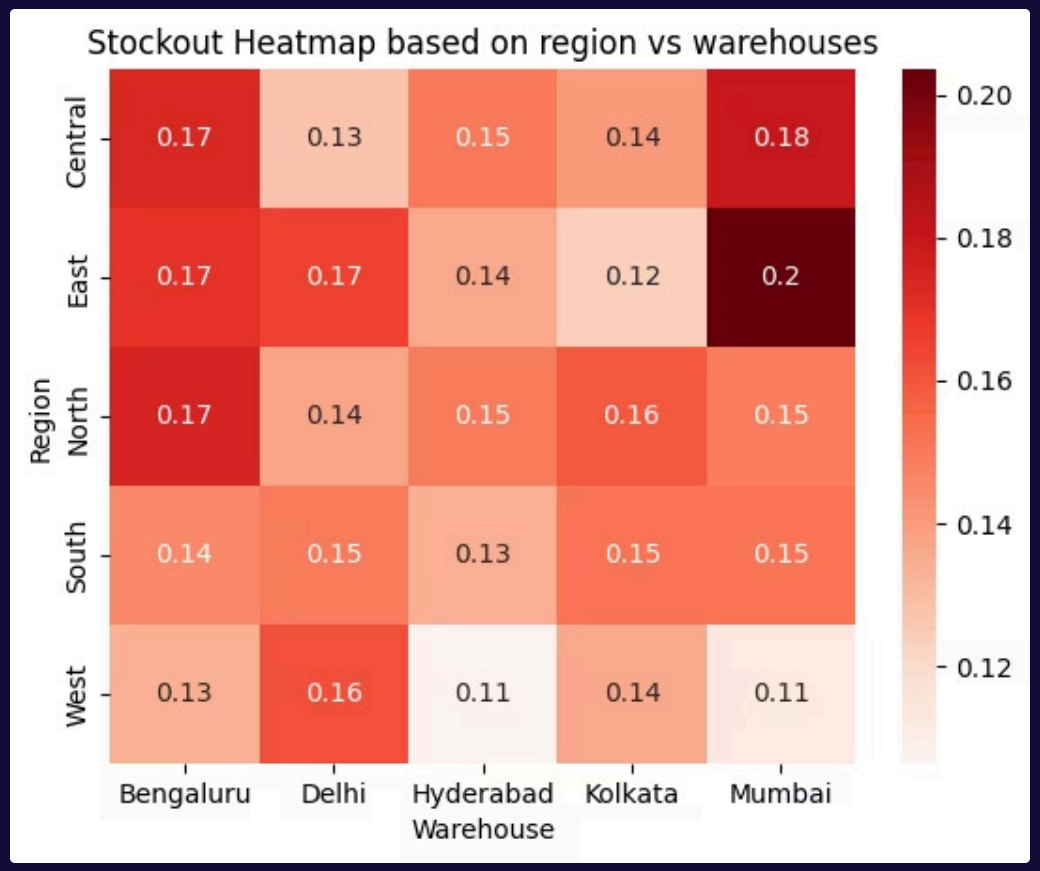
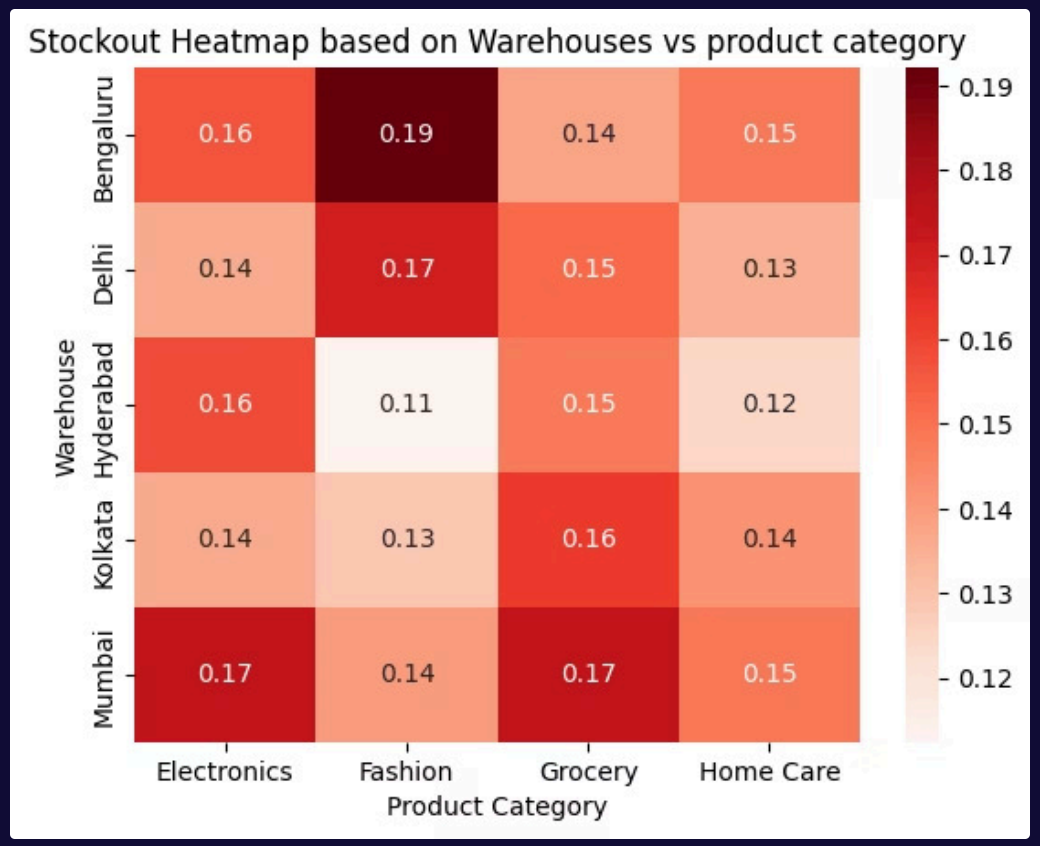
Supply Chain Optimization: Data-Driven Solutions for Stockout Prevention

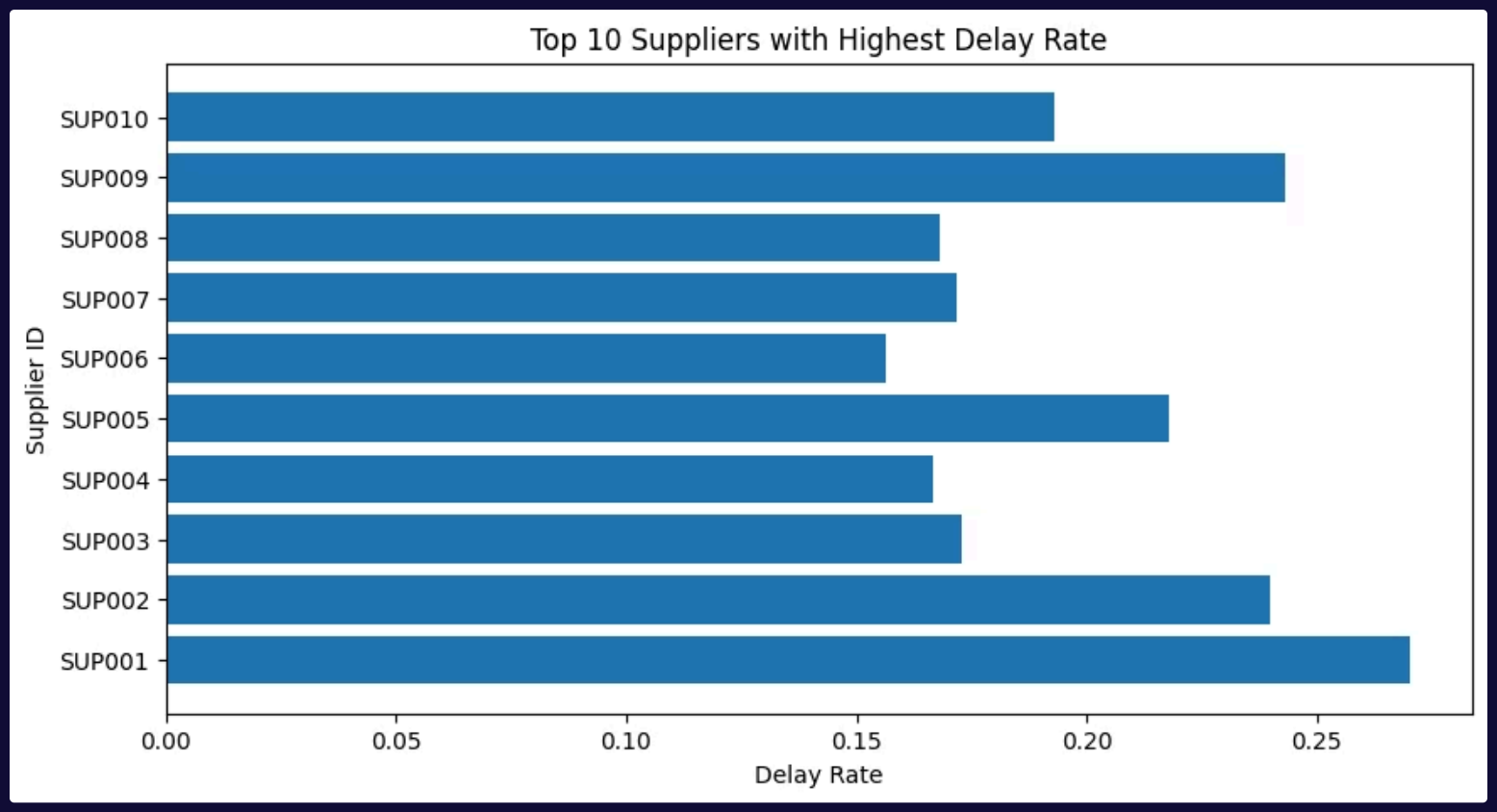
Addressing stockout challenges through regional inventory strategies and predictive modeling for enhanced operational efficiency.





- The warehouses which are getting stocked out in specific region supplies like the Mumbai warehouse in the eastern front is over stocked out so in such situations the closer or the stock full warehouses like Kolkata should be compiled.
- Stockouts in Fashion products form warehouse like Bengaluru and electronics product in Mumbai warehouse are high, so the supply should be distributed among other warehouses to maintain uniformity and less stockout issues.

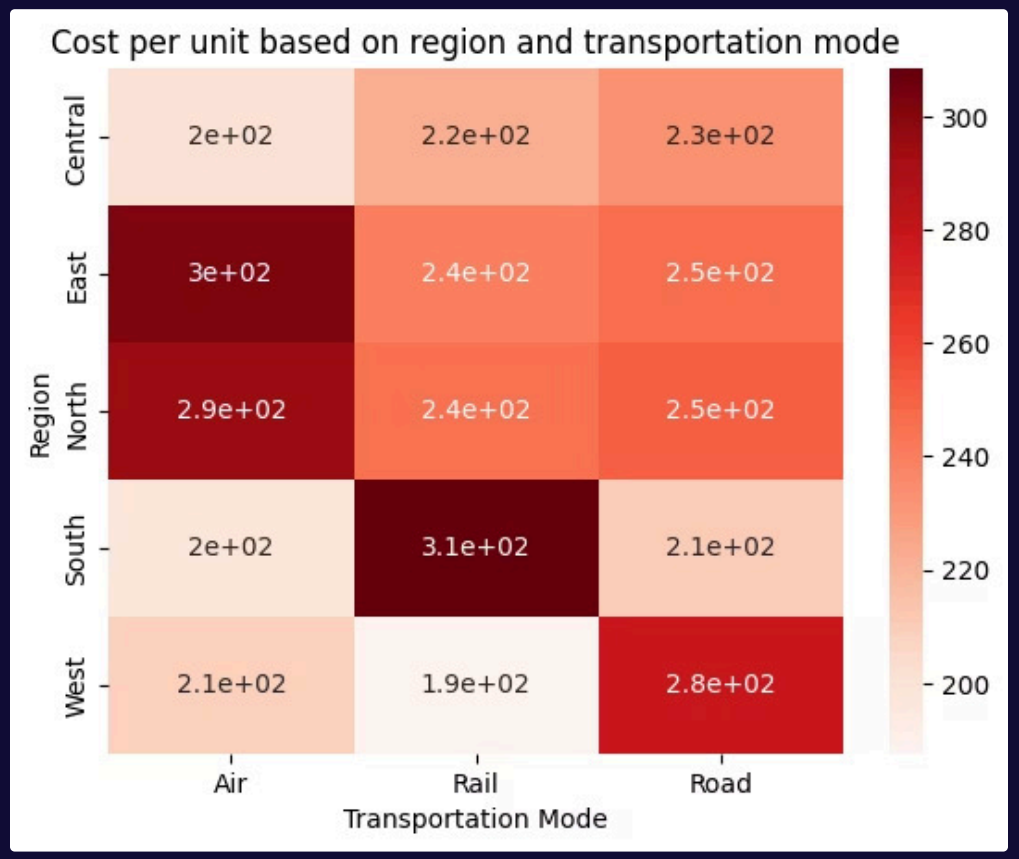
Delays due to Suppliers' Inefficiencies and its Management

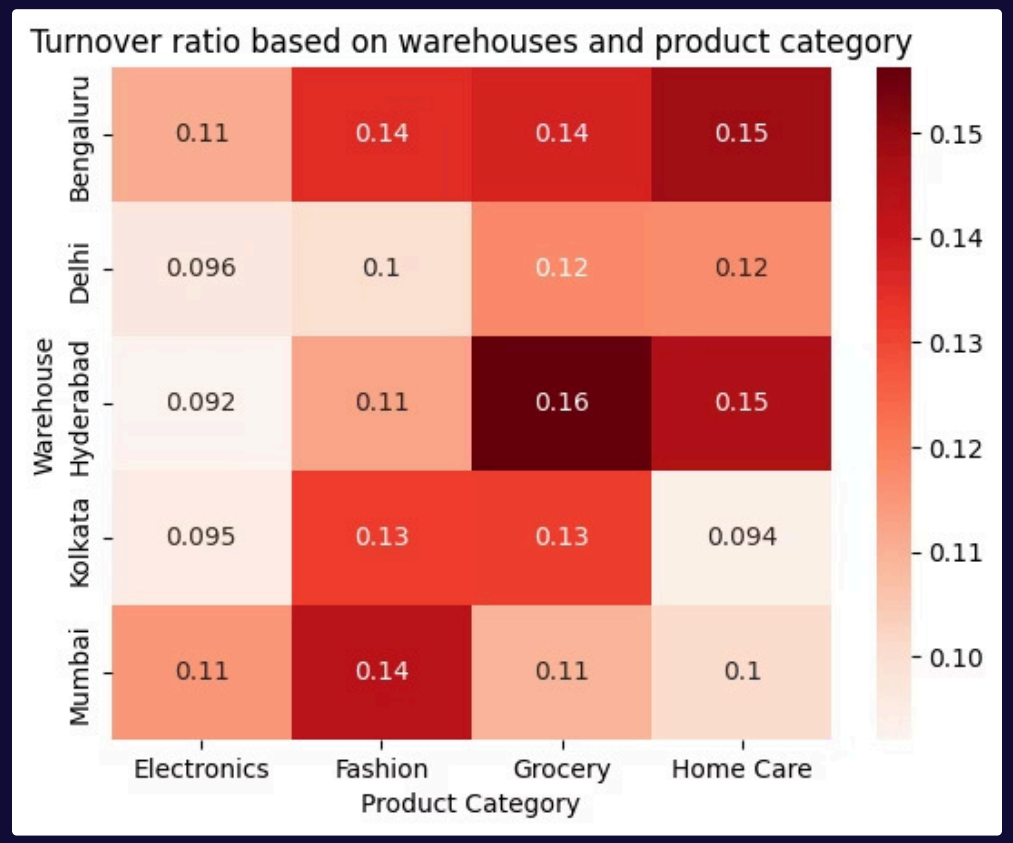


S.No.	Supplier ID	Order Lead Time Days	Fill rate	Delay Flag
42	SUP043	8.160494	1.257140	0.234568
8	SUP009	7.495495	1.564661	0.243243
37	SUP038	7.673469	1.570017	0.234694
29	SUP030	7.571429	1.603985	0.197802
31	SUP032	7.693069	1.636148	0.168317

- The Suppliers with mean delay rate more than 0.22 are bulked with orders or are inefficient in providing the supplies to the warehouses. So, either the supplies should be distributed among the other distributers or orders from such suppliers should be reduced or cutoff.
- Supplier delay issues: 10 suppliers with >0.22 delay rate

Transportation and Holding Cost management and Strengthening

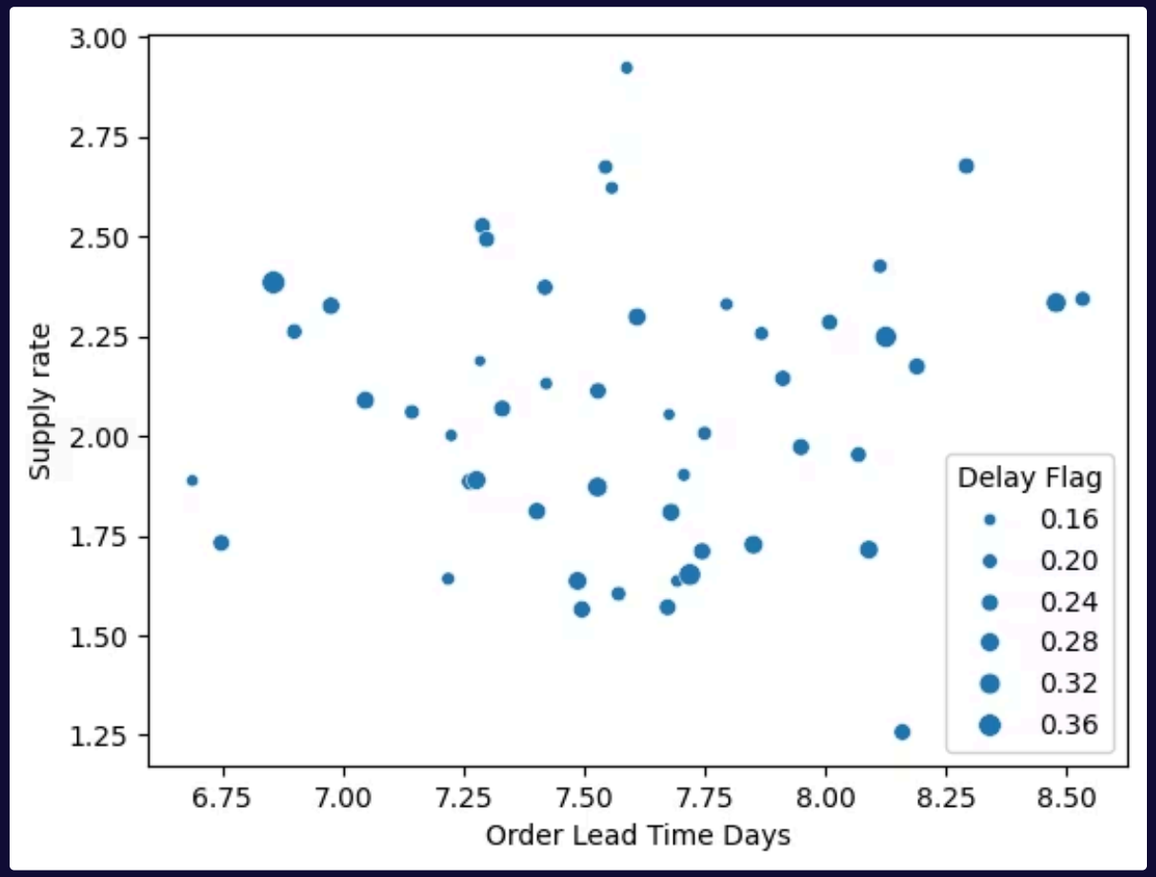




- The Heatmap based on region and transportation mode where the cost per unit is very high should be parsed in other mode of transportation specifically for the corresponding regions. e.g The deliveries through rail to south region is very high so it should be shifted to some other modes.
- For specific product categories like Electronics in Hyderabad Warehouse and Home Care in Kolkata Warehouse, the turnover ratio is way too low that implies the holding cost in these region is high. On the basis of supply and demand the stock of these products in the warehouses should be limited in order to reduce the holding cost.

Supplier Performance Analysis: Strengthening the Chain

Identifying and addressing underperforming suppliers is vital for a resilient supply chain.



n the Scatter plot, we can compare the fill rate with order lead time days based on the delays.From the scatter plot, we can draw a conclusion that some suppliers are inefficient in a way that the order lead time days and delays both are high with a subsequent low fill rate.

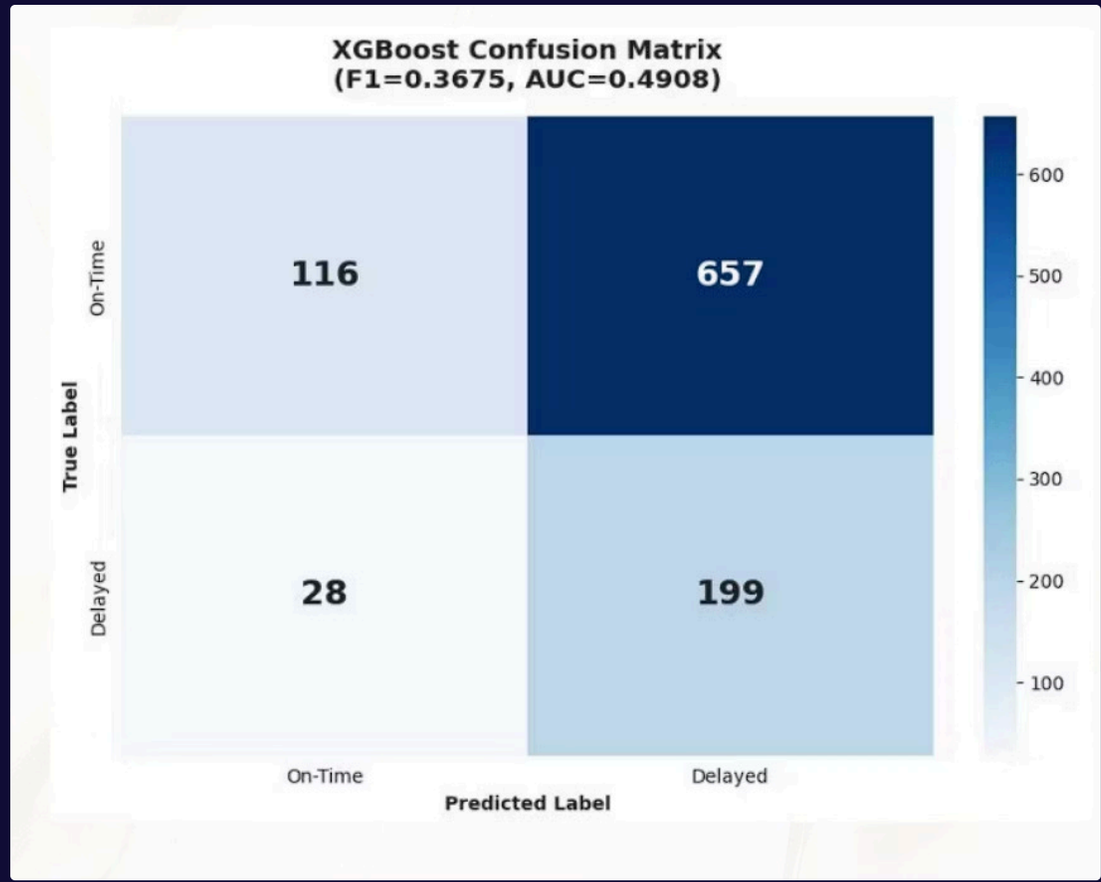
- **Recommendation:**
  - Reduce or eliminate bottom 15-20% of inefficient suppliers from the plot.
  - Distribute orders among more reliable suppliers to mitigate risk.

Predictive Model Results: Early Warning System

Our XGBoost model provides critical insights for proactive delay management as an early indicator.

Key Outcomes

- True Positives: 199 delayed orders correctly identified from the confusion matrix.
- False Negatives: Only 28 missed delays, ensuring high reliability of the prediction model.
- With the above 199 True positives and only 28 false negatives the model captures the majority of the delay cases indicating high recall rate.
- It shifts the supply chain from reactive to a proactive based model enabling preventive actions like Rerouting, Supplier Escalation and Safety Stock activation.



Strategic Recommendations & Action Plans

A phased approach to implement data-driven optimizations for a robust supply chain.

01

Implement Inventory Redistribution

Deploy region-specific strategies to balance stock levels effectively as suggested.

02

Optimize Transportation Modes

Shift to cost-efficient modes based on in-depth analysis.

03

Consolidate Supplier Base

Eliminate underperforming suppliers and strengthen partnerships with reliable vendors.

04

Deploy Predictive Model

Models like the XGBoost model used above can be helpful in real-time delay detection and mitigation.

05

Establish Cross-Regional Coordination

Develop protocols for seamless warehouse management across all regions.